PythFLOW

Litepaper

The No-Code Frontier for DeFi Composability on Cardano

Design. Deploy. Democratize.

1. Executive Summary: The Vision

The Decentralized Finance (DeFi) ecosystem has grown in complexity, leaving behind a vast majority of users and professionals who possess the financial vision but lack the technical mastery of complex programming languages.

PythFLOW was built to close this gap. Our vision is to democratize access to high-level financial infrastructure, enabling any individual to design, deploy, and manage DeFi protocols with institutional precision through a visual interface powered by Artificial Intelligence.

We are not just building a tool; we are building the bridge toward the democratization of on-chain logic. **Iron Pig** is the living proof of this vision: the first institutional-grade financial protocol built entirely on the PythFLOW platform.

2. The Problem: Technical Barriers to Entry

Today, deploying a functional protocol on Cardano requires deep expertise in:

- **Aiken or Plutus:** Smart contract languages with steep learning curves.
- **Oracle Infrastructure:** Manual integration of external data feeds.
- **Transaction Management:** Complex handling of eUTXO and data serialization.

These barriers confine innovation to a small circle of developers, excluding economists, risk analysts, creative entrepreneurs, and the general public. The result is a fragmented ecosystem where financial creativity remains locked behind an insurmountable technical wall.

3. The Solution: PythFLOW

PythFLOW is a revolutionary infrastructure platform that transforms smart contract creation into an intuitive drag-and-drop process.

Technology Pillars

- **Aiken:** We leverage the efficiency and security of Aiken to generate the contract backend.
- **MeshJS:** We facilitate seamless frontend interaction and wallet management.
- **Generative AI:** An AI agent acts as a co-pilot, translating natural language intentions into logical node connections.

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4. Data Architecture: Pyth Network Integration

The precision of a DeFi protocol depends on the quality of its data. PythFLOW uses **Pyth Network** as its primary oracle layer, providing real-time price data with sub-second latency.

Nodes within the workflow can:

1. **Price Ingestion:** Read price feeds directly from Pyth.
2. **Conditional Logic:** Define market-data-based triggers (e.g., "If ADA price < X, execute liquidation").
3. **Native Integration:** Embed oracle data directly into the Aiken contract validator with no manual friction.

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5. The Flow: From Idea to Mainnet

The creation process in PythFLOW is built on an infinite canvas where financial creativity comes to life:

4. **Blank Canvas:** The user starts with a canvas where they can add nodes manually or ask the AI to generate a base structure.
5. **Node Configuration:** Each node represents a function (SWAP, LEND, ORACLE, MINT).
6. **Logic Generation:** The system translates the visual graph into robust, verified Aiken code.
7. **Instant Deployment:** With a single click, the transaction is submitted to deploy the contract on the network.

6. Iron Pig: The Flagship Vertical

IRON PIG

Lending & Collateral Management Protocol built on PythFLOW

What is Iron Pig?

Iron Pig is the first institutional-grade DeFi protocol developed entirely using the PythFLOW platform. It serves as both a living proof of concept and a standalone product: a collateralized lending system on Cardano that proves on-chain complexity can be built without writing a single line of code.

Iron Pig leverages the native integration with **Pyth Network** to calculate collateralization ratios in real time, trigger automatic liquidations, and manage liquidity pools with a level of data precision that previously required specialized engineering teams.

Iron Pig Functional Architecture

Module	Description
COLLATERAL NODE	Accepts ADA and native Cardano tokens as collateral. Calculates Loan-to-Value (LTV) in real time using Pyth price feeds.
LENDING POOL NODE	Manages liquidity pools for lenders. Distributes proportional yields based on capital provided.
LIQUIDATION TRIGGER	Automatic collateral ratio monitor. If LTV exceeds the defined threshold, it triggers liquidation without human intervention.
ORACLE INTEGRATION	Direct connection to Pyth Network for sub-second pricing of ADA, BTC, ETH, and Cardano ecosystem tokens.
RISK PARAMETERS NODE	Allows visual configuration of risk parameters: maximum LTV, liquidation penalty, and grace period.

Operation Flow

The lifecycle of a loan on Iron Pig operates as follows:

8. **Collateral Deposit:** The borrower deposits ADA or other supported assets into the COLLATERAL node. The value is calculated instantly using the Pyth feed.
9. **Loan Request:** The protocol determines the maximum borrowable amount based on the configured LTV and issues the assets to the borrower.
10. **Continuous Monitoring:** The LIQUIDATION TRIGGER monitors the collateral ratio on

every block. If the collateral price drops, the system reacts automatically.

11. **Repayment or Liquidation:** The borrower can repay at any time to recover their collateral, or the protocol executes liquidation to protect lenders.

Iron Pig as Proof of PythFLOW's Power

Why does Iron Pig validate the PythFLOW vision?

- **Built without code:** The entire protocol was designed visually on the PythFLOW canvas, without writing Aiken manually.
- **Real data, real decisions:** Pyth Network prices feed directly into the contract logic in real time.
- **Composability demonstrated:** The COLLATERAL, LENDING, and ORACLE nodes were visually interconnected in minutes.
- **Deployed on Testnet:** Iron Pig operates on the Cardano testnet, validating PythFLOW's full end-to-end cycle.

Iron Pig Roadmap

Phase	Milestone	Description
Phase 1	**Testnet Launch**	Iron Pig live on the Cardano testnet. Full cycle validation: collateral → loan → liquidation.
Phase 2	**Security Audit**	Independent review of Aiken code generated by PythFLOW. AI validation of all protocol nodes.
Phase 3	**Mainnet Beta**	Launch with initial liquidity pools, conservative LTV, and position caps for risk management.
Phase 4	**Asset Expansion**	Support for additional native Cardano tokens. Integration of Pyth feeds for new trading pairs.
Phase 5	**Governance**	Transition to decentralized protocol governance via the PythFLOW ecosystem native token.

7. Execution Environment & Security

With rapid iteration and user safety in mind, PythFLOW includes an execution environment designed to facilitate the development, testing, and safe deployment of protocols like Iron Pig:

- **Native Testnet Support:** Optimized for the Cardano test network, allowing experimentation with no real capital at risk.
- **Integrated Burner Wallet:** A temporary (disposable) in-browser wallet for quick interactions.
- **Direct Faucet:** Built-in button to obtain test funds (tADA) instantly.
- **AI Validation:** The agent reviews node logic to detect potential security errors before deployment.
- **Risk Parameter Auditing:** For verticals like Iron Pig, the AI validates that LTV and liquidation thresholds are consistent with market best practices.

8. PythFLOW Verticals Ecosystem

Iron Pig is the first vertical in the ecosystem, but not the last. PythFLOW's modular architecture is designed to power multiple specialized DeFi protocols, each built on the same base platform. Any team or individual can design their own vertical using PythFLOW:

PythFLOW Ecosystem Verticals

- **Iron Pig (Lending):** Collateralized loans with automated risk management. [LIVE ON TESTNET]
- **PythSWAP (DEX):** Decentralized exchange with oracle-based pricing powered by Pyth Network.
- **PythYIELD (Yield):** Yield optimizer that automatically redirects liquidity based on market rates.
- **PythSTABLE (Stablecoin):** On-chain collateral-backed stablecoin issuance with automated liquidations.
- **Custom Protocol:** Any team can design and deploy their own DeFi protocol using the PythFLOW canvas.

9. Conclusion

PythFLOW does not just simplify development; it redefines who can be a financial architect. By combining the robustness of Cardano's eUTXO model with the speed of Pyth data and the accessibility of No-Code tools, we are ready to unleash the next wave of innovation in the crypto ecosystem.

Iron Pig embodies this philosophy: a robust, functional lending protocol built without a single line of hand-written code, deployed on Cardano, powered by real-time Pyth data. It is the definitive proof that the future of DeFi is accessible to everyone.

PythFLOW × Iron Pig

Building the next generation of DeFi infrastructure on Cardano.